

DBSE WEEKLY LAB EXPERIMENTS

KAIRAV VASHI

2520030582

SECTION – 7

CSE

WEEK-2

The screenshot shows the SQL Developer interface with a script titled 'SQL File 3'. The script contains the following SQL commands:

```
1 CREATE DATABASE IF NOT EXISTS bookflow_db;
2 USE bookflow_db;
3
4 DROP TABLE IF EXISTS Donation_History;
5 DROP TABLE IF EXISTS Loans;
6 DROP TABLE IF EXISTS Members;
7 DROP TABLE IF EXISTS Books;
8
9 CREATE TABLE Books (
10     book_id INT PRIMARY KEY,
11     title VARCHAR(100) NOT NULL,
```

The 'Result Grid' shows the execution of the first two commands:

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949

The 'Output' pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	19:47:49	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected	0.000 sec
2	19:47:49	USE bookflow_db	0 row(s) affected	0.000 sec

The screenshot shows the SQL Developer interface with a script titled 'SQL File 3'. The script contains the following SQL commands:

```
9 CREATE TABLE Books (
10     book_id INT PRIMARY KEY,
11     title VARCHAR(100) NOT NULL,
12     isbn VARCHAR(20) UNIQUE,
13     published_year INT CHECK (published_year < 2027)
14 );
15
16 INSERT INTO Books (book_id, title, isbn, published_year) VALUES
17 (1, 'The Great Gatsby', '9780743273565', 1925),
18 (2, 'To Kill a Mockingbird', '9780061120084', 1960),
19 (3, '1984', '9780451524935', 1949);
```

The 'Result Grid' shows the execution of the first two commands:

book_id	title	isbn	published_year
1	The Great Gatsby	9780743273565	1925
2	To Kill a Mockingbird	9780061120084	1960
3	1984	9780451524935	1949

The 'Output' pane shows the execution results:

#	Time	Action	Message	Duration / Fetch
1	19:47:49	CREATE DATABASE IF NOT EXISTS bookflow_db	1 row(s) affected	0.000 sec
2	19:47:49	USE bookflow_db	0 row(s) affected	0.000 sec
3	19:47:49	DROP TABLE IF EXISTS Donation_History	0 row(s) affected. 1 warning(s): 1051 Unknown table 'bookflow_db.donation_history'	0.000 sec
4	19:47:49	DROP TABLE IF EXISTS Loans	0 row(s) affected. 1 warning(s): 1051 Unknown table 'bookflow_db.loans'	0.000 sec
5	19:47:49	DROP TABLE IF EXISTS Members	0 row(s) affected. 1 warning(s): 1051 Unknown table 'bookflow_db.members'	0.000 sec
6	19:47:49	DROP TABLE IF EXISTS Books	0 row(s) affected. 1 warning(s): 1051 Unknown table 'bookflow_db.books'	0.000 sec
7	19:47:49	CREATE TABLE Books (book_id INT PRIMARY KEY, title VARCHAR(100) NOT NULL, isbn VARCHAR(20) UNIQUE, published_year INT CHECK (published_year < 2027))	0 row(s) affected	0.032 sec
8	19:47:49	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (1, 'The Great Gatsby', '9780743273565', 1925), (2, 'To Kill a Mockingbird', '9780061120084', 1960), (3, '1984', '9780451524935', 1949);	3 row(s) affected. Records: 3 Duplicates: 0 Warnings: 0	0.000 sec

Navigator: SCHEMAS
Filter objects
sys

SQL File 3*

```

21 SELECT * FROM Books;
22
23 CREATE TABLE Members (
24     member_id INT PRIMARY KEY,
25     full_name VARCHAR(100),
26     email VARCHAR(100) UNIQUE
27 );
28
29 INSERT INTO Members (member_id, full_name, email) VALUES
30 (101, 'Kairav', 'kairav1@email.com'),
31 (102, 'Kairav', 'kairav2@email.com'),
32 (103, 'Kairav', 'kairav3@email.com');

```

Result Grid

member_id	full_name	email
101	Kairav	kairav1@email.com
102	Kairav	kairav2@email.com
103	Kairav	kairav3@email.com

Books 21 Members 22 Loans 23 Result 24 Result 25 Books 26

Output

Action Output

#	Time	Action	Message	Duration / Fetch
9	19:47:49	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
10	19:47:49	CREATE TABLE Members (member_id INT PRIMARY KEY, full_name VARCHAR(100), email VARCHAR(100) UNIQUE)	0 row(s) affected	0.016 sec
11	19:47:49	INSERT INTO Members (member_id, full_name, email) VALUES (101, 'Kairav', 'kairav1@email.com'), (102, 'Kairav', 'kairav2@email.com'), (103, 'Kairav', 'kairav3@email.com')	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.016 sec
12	19:47:49	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
13	19:47:49	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE)	0 row(s) affected	0.031 sec
14	19:47:49	INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'), (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'), (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12')	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec

Object Info Session

Navigator: SCHEMAS
Filter objects
sys

SQL File 3*

```

33 SELECT * FROM Members;
34
35
36 CREATE TABLE Loans (
37     loan_id INT PRIMARY KEY,
38     member_id INT,
39     book_id INT,
40     loan_date DATE,
41     FOREIGN KEY (member_id) REFERENCES Members(member_id),
42     FOREIGN KEY (book_id) REFERENCES Books(book_id)
43 );
44

```

Result Grid

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12

Books 21 Members 22 Loans 23 Result 24 Result 25 Books 26

Output

Action Output

#	Time	Action	Message	Duration / Fetch
13	19:47:49	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE)	0 row(s) affected	0.031 sec
14	19:47:49	INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1, '2025-01-05'), (2, 102, 2, '2025-01-08'), (3, 103, 3, '2025-01-10'), (4, 101, 2, '2025-02-01'), (5, 102, 1, '2025-02-05'), (6, 103, 2, '2025-02-12')	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
15	19:47:49	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
16	19:47:49	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.member_id = m.member_id INNER JOIN Books b ON l.book_id = b.book_id	10 row(s) returned	0.000 sec / 0.000 sec
17	19:47:49	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year ORDER BY published_year	3 row(s) returned	0.000 sec / 0.000 sec
18	19:47:49	CREATE TABLE Donation_History (donation_id INT PRIMARY KEY, book_id INT, donor_name VARCHAR(100), donation_date DATE)	0 row(s) affected	0.031 sec

Object Info Session

Navigator
SCHEMAS
Filter objects
sys

Administration Schemas
Information
No object selected

SQL File 3*
Limit to 1000 rows

```

45 INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES
46 (1, 101, 1, '2025-01-05'),
47 (2, 102, 2, '2025-01-08'),
48 (3, 103, 3, '2025-01-10'),
49 (4, 101, 2, '2025-02-01'),
50 (5, 102, 1, '2025-02-05'),
51 (6, 103, 2, '2025-02-12'),
52 (7, 101, 3, '2025-03-01'),
53 (8, 102, 3, '2025-03-07'),
54 (9, 103, 1, '2025-03-15'),
55 (10, 101, 1, '2025-04-01');
56

```

Result Grid
loan_id member_id book_id loan_date

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12

Books 21 Members 22 Loans 23 Result 24 Result 25 Books 26

Output
Action Output

#	Time	Action	Message	Duration / Fetch
13	19:47:49	CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DAT...	0 row(s) affected	0.031 sec
14	19:47:49	INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES (1, 101, 1, '2025-01-05'), (2, 102, 2, '...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.000 sec
15	19:47:49	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
16	19:47:49	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.m...	10 row(s) returned	0.000 sec / 0.000 sec
17	19:47:49	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year ORDE...	3 row(s) returned	0.000 sec / 0.000 sec
18	19:47:49	CREATE TABLE Donation_History (donation_id INT PRIMARY KEY, book_id INT, donor_name VAR...	0 row(s) affected	0.031 sec

Object Info Session

Navigator
SCHEMAS
Filter objects
sys

Administration Schemas
Information
No object selected

SQL File 3*
Limit to 1000 rows

```

57 SELECT * FROM Loans;
58
59 SELECT m.full_name AS Member_Name, b.title AS Book_Title
60 FROM Loans l
61 INNER JOIN Members m ON l.member_id = m.member_id
62 INNER JOIN Books b ON l.book_id = b.book_id;
63
64 SELECT published_year, COUNT(book_id) AS Total_Books
65 FROM Books
66 GROUP BY published_year
67 ORDER BY published_year;
68

```

Result Grid
loan_id member_id book_id loan_date

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12

Books 21 Members 22 Loans 23 Result 24 Result 25 Books 26

Output
Action Output

#	Time	Action	Message	Duration / Fetch
15	19:47:49	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
16	19:47:49	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.m...	10 row(s) returned	0.000 sec / 0.000 sec
17	19:47:49	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year ORDE...	3 row(s) returned	0.000 sec / 0.000 sec
18	19:47:49	CREATE TABLE Donation_History (donation_id INT PRIMARY KEY, book_id INT, donor_name VAR...	0 row(s) affected	0.031 sec
19	19:47:50	START TRANSACTION	0 row(s) affected	0.000 sec
20	19:47:50	INSERT INTO Books (book_id, title, isbn, published_year) VALUES (4, 'Animal Farm', '9780451526342', 1945)	1 row(s) affected	0.000 sec

Object Info Session

Navigator
SCHEMAS
Filter objects
sys

Administration Schemas
Information
No object selected

SQL File 3*
Limit to 1000 rows

```

57 SELECT * FROM Loans;
58
59 SELECT m.full_name AS Member_Name, b.title AS Book_Title
60 FROM Loans l
61 INNER JOIN Members m ON l.member_id = m.member_id
62 INNER JOIN Books b ON l.book_id = b.book_id;
63
64 SELECT published_year, COUNT(book_id) AS Total_Books
65 FROM Books
66 GROUP BY published_year
67 ORDER BY published_year;
68

```

Result Grid
published_year Total_Books

published_year	Total_Books
1925	1
1949	1
1960	1

Books 21 Members 22 Loans 23 Result 24 Result 25 Books 26

Output
Action Output

#	Time	Action	Message	Duration / Fetch
15	19:47:49	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
16	19:47:49	SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.m...	10 row(s) returned	0.000 sec / 0.000 sec
17	19:47:49	SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year ORDE...	3 row(s) returned	0.000 sec / 0.000 sec

Read Only